

Autarco

RS485_MODBUS Communication Protocol

Ver 17

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Table of content

1. Overview	2
2. Physical interface	2
3. Communication Description	2
3.1. Data type	2
3.2. Inter-frame interval requirement	2
3.3. Data frame	2
4. Error information and data process	3
5. Detail description of the Protocol	4
5.1. Inverter Model Parameter Address Definition	4
5.2. Register address of inverter operation information.	5
5.3. Register address of inverter setting	17
5.4. PELD (external device) operating information.....	33
5.5. PELD (External device) setting	35
Example	36
6.1. 04 function code	36
6.2. 03 function code	36
6.3. 06 function code	37
6.4. 10 function code	37
6.5. Broadcast setting	37
Appendix 1:	38
Appendix 2:	41

1. Overview

This protocol adopted MODBUS RTU regulation, is applicable to the communication protocol between Autarco grid-tied inverter and PC monitoring software. This Protocol can read operational information and control the inverter in real time.

2. Physical interface

Adopts RS485 Receiver-Transmitter, Client-Server Model:

Slave Address 1~247

Baud rate 9600bps
Parity checking None
Data 8
Stop 1
model RTU

3. Communication Description

3.1. Data type

U16: 2-byte unsigned integer, high first and then low;

S16: 2-byte signed integer, high first and then low;

U32: 4-byte unsigned integer, the high byte is in front of the low byte, and the high word is in the front and the low word is behind;

S32: 4-byte signed integer, the high byte is in front of the low byte, and the high word is in the front and the low word is behind;

3.2. Inter-frame interval requirement

More than 300ms communications frame interval is required. Recommended max data frame 100 bytes (50 registers)

3.3. Data frame

Slave Address	Function code	Data	CRC Check
8-Bits	8-Bits	Nx8-Bits	16-Bits

Slave Address: Is the corresponding slave address, it must be match with inverter address.

Broadcast Address 0xFF, only used for remote control

Function code: 03H, 04H, 06H and 10H are available

Function code(Hex)	Name	Reg Address	Function
03H	Read the holding registers	40001-49999	Read the setting content of holding registers
04H	Read the input registers	30001-39999	Read the detail information of the inverter
06H	Write a single holding registers	40001-49999	Set single-byte functions
10H	Write multiple holding registers	40001-49999	Set multi-byte function

Data: Including the start register address, data length, the number of data bytes, data content. 02H low-byte and follow high-byte, others high-byte first, and follow by low byte.

CRC Check: CRC look-up table checking mode. High-byte first, and follow by low Byte.

When the slave device receives address 0xFF and the function code is “write”, receive the command but not response.

4. Error information and data process

Slave Response (Hex)

Slave Address	Function code	Error code	CRC Check	
xx	xx 0x80	xx	Low byte	High byte
			xx	xx

When the inverter communication module detected an error other than CRC error, it must response to the master device. (High byte of function code is 1 which is adding 128 to the function code) **Inverter**

com module response to the Error Code

0x01 illegal function code, the server doesn't understand the function code

0x02 illegal data address, in relation to requests

0x03 illegal data, in relation to requests.

0x04 Service failure, Inverter com module can't get access to the data during execution

0x05 HMI and Dsp communication failure;

5. Detail description of the Protocol

5.1. Inverter Model Parameter Address Definition

Corresponding function code is 0x04. The following table has the same address with the actual address of the message frame. No need extra offset or transform

Register address (Decimal)	Name	Data type	Remark
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35000	Autarco inverter type definition	U16	definition: 0000---no definition 1010---1phase inverter(For: 0.7-8K, 7-10K) 1020---3 phase inverter(For 5-20K 3P) 1021---3 phase inverter(For 25-50K/50-70K/80-110K/90-136K/125K/255K) 2030--- 1 phase low voltage energy storage inverter 2031--- 1 phase low voltage AC Couple energy storage inverter 2040---1 phase high voltage energy storage inverter 2050--- 3 phases low voltage energy storage inverter 2060--- 3 phases high voltage energy storage inverter LH 2070---3 phases high voltage energy storage inverter LH-MII 2080---1 phase high voltage energy storage inverter MH-MII 1070---external PELD 3010---OGI OFF GRID INVERTER description: 1、high 8 bit means protocol version, low 8 bit means inverter model 10: see<RS485_MODBUS(INV-3000ID PELD-36000ID)inverter protocol>; 20: see<RS485_MODBUS(ESINV-33000ID)energy storage inverter protocol> 30: see< off-grid FreeModbus> 2、0000H-no definition model。 Master device can choose the device UI of related model 3、this address will not be limited with wake up time,it can get address information as soon as LCD is powered up
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5.2. Register address of inverter operation information.

Corresponding function code is 0x04. The function code is 0x04, the register address needs to offset one bit.

Note: METER SN number is represented by the SN number of the inverter plus the slave address of the inverter to indicate the SN number of the METER, which is used to distinguish whether the device has an electric meter device. For single phase, the data of phase A shall prevail. The positive value of METER active power means sending electricity to the grid, and the negative number means taking electricity from the grid.

Register	name	Data type	Unit	Remark	Note
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address					
3000	Product model	U16		Hex	
3001	DSP software version	U16		Hex	
3002	Reserved				
3003	AC output type	U16		Single 3P4Wires 3P3Wires 3P4Wires or 3P3Wires both available	
3004	DC input type	U16		0-1 DC input 1-2 DC input 2-3 DC input 3-4 DC input 19-20 DC input	
3005-3006	Active power	S32	1W		
3007-3008	Total DC output power	U32	1W		
3009-3010	Total energy	U32	1kWh	If there is a need to display the decimal part, it can be associated with the 3117 register address here	
3011-3012	Energy this month	U32	1kWh		
3013-3014	Energy last month	U32	1kWh		
3015	Energy today	U16	0.1kWh		
3016	Reserve	U16	0.1kWh		
3017-3018	Energy this year	U32	1kWh		
3019-3020	Energy last year	U32	1kWh		

3021	HMI version	U16			
3022	DC voltage 1	U16	0.1V		Applicable to inverters: mainly 1. 0.7-10K single phase 2. 5-25K series 3. 25-50K series
3023	DC current 1	U16	0.1A		
3024	DC voltage 2	U16	0.1V		
3025	DC current 2	U16	0.1A		
3026	DC voltage 3	U16	0.1V		
3027	DC current 3	U16	0.1A		
3028	DC voltage 4	U16	0.1V		
3029	DC current 4	U16	0.1A		
3030	Alarm code data	U16		In conjunction with the 3044 register address, it is used for subdivided fault information display.	Example: 3044 register read information is 1020, The 3030 register is one of 0000-0004, corresponding to the display
3031	Initialize ground voltage Value	U16	0.1V		
3032	DC busbar voltage	U16	0.1V		
3033	DC half-busbar voltage	U16	0.1V		
3034	AB line voltage / A phase voltage	U16	0.1V	Output type (3003) is 1: Output phase voltage 2: Output line voltage	For the AC side voltage, currently only the 230K series is line voltage, and all other systems are phase voltage
3035	BC line voltage / B phase voltage	U16	0.1V	Output type (3003) is 1: Output phase voltage 2: Output line voltage	
3036	CA line voltage / C phase voltage	U16	0.1V	Output type (3003) is 1: output phase voltage 2: Output line voltage 0: Output single-phase voltage	
3037	A phase current	U16	0.1A		
3038	B phase current	U16	0.1A		

3039	C phase current	U16	0.1A		
3040	Reserved				
3041	Standard working mode	U16		working mode : 00---No response mode 01---Volt-watt default 02---Volt-var 03---Fixed power factor 04---Fix reactive power 05---Power-PF 06---Rule21 Volt-watt	
3042	Inverter temperature	S16	0.1°C		Note : AC NTC (IGBT)

3043	Grid Frequency	U16	0.01Hz		
3044	Inverter status	U16		See Appendix 1	
3045-3046	Limited active power adjustment rated power output value	S32	1W	Switch : 0xAA: Value : [06 Code Address 3052] × Rated Power Switch : 0x55: Value : Rated Power	Note : This switch is in function code 06 Address 3070
3047-3048	Reactive power regulation rated power output value	S32	1Var	Switch : 0x55: Value 0; Switch : 0xA1: Value : [06 Code Address 3051] × Rated Power Switch: 0xA2: Value: $\sqrt{1^2 - [06 \text{ Code Address } 3054]^2} \times \text{Rated Power}$	Note : This switch is in function code 06 Address 3071
3049	Inverter control word	U16			
3050	Actual value of limited Active power	U16	1%	10000<-->100% Setting range (0-100%) 100% refers to the nominal nominal power.	

3051	Actual adjust value of power factor	S16	0.001	Low digits of power factor: (800<-->0.80, 1000<-->1.00) (-800<-->-0.80 , -1000<-->-1.00) (Power factor 1.00 and -1.00 are the same)	Setting range (-1~-0.80、0.80~1)
3052	Actual power factor adjustment	S16	0.001	Low digits of power factor : (800<-->0.80, 1000<-->1.00) (-800<-->-0.80, -1000<-->-1.00) (Power factor 1.00 and -1.00 are the same)	Setting range (-1~-0.8、0.8~1) This feature is only for the fixed power factor setting function of standard mode 3.
3053	Reactive power value	S16	1%	10000<-->100% Setting range (-6000- +6000) default :0	This function is only for the reactive power setting of standard mode 4.
3054	Standard	U16		See Appendix 2	
3055	Power curve number	U16			
3056-3057	Reactive power value	S32	1Var		
3058-3059	Apparent power value	S32	1VA		
3060	Real-time power factor	S16	0.001		Only Available for 50-70K/255K Series
3061	Inverter serial number SN_1	U16		Inverter sequence (hexadecimal display) Example:	

3062	Inverter serial number SN_2	U16		The numerical upload value of 3061 is 0x4321 The numerical upload value of 3062 is 0x8765. The numerical upload value of 3063 is 0xCBA9 The numerical upload value of 3064 is 0x0FED Sn number: 12345679ABCDEF	
3063	Inverter serial number SN_3	U16			
3064	Inverter serial number	U16			

	SN_4				
3065-3072	Reserved				
3073	System time - year	U16		0-99	
3074	System time - month	U16			
3075	System time - day	U16			
3076	System time-hour	U16			
3077	System time-minute	U16			
3078	System time - second	U16			
3079	DRM	U16		DRM Number	
3080-3085	Reserved				
3086	Leakage current protection value	U16	1mA	1<-->1mA	
3087	Insulation resistance protection value	U16	1kOhm	1<-->1k	
3088	Reserved				
3089	Reserved				
3090	Power limit switch	U16		0xAA Enable limit power switch, 0x55 is the power limit switch (restrict power recovery to 100%)	Note: Work with Code 06 Address 3070
3091	Reactive power switch	U16		0x55 Off, power factor recovery 1, reactive power ratio recovery 0; 0xA1 reactive power ratio setting is valid; The 0xA2 power factor 02 setting is valid.	Note: Work with Code 06 Address 3071
3092	Real Time PV insulation resistance	U16	1kΩ	Range:0-32000kΩ	
3093	Inverter Temperature	S16	0.1°C		

3094	Inverter installation Method	U16		Inverter installation method (default 0; 0 is 90° installation method, 1 is 15° installation method; according to different installation methods, the inverter implements different temperature derating logic)	
3095-3105	Reserve				

3106	Ileak Real time data	U16	1mA	1<-->1mA	
3107-3110	Reserve	U16			
3111	Internal PELD Switch (PELD Soft Switch for AU 2020 Code)	U16		1、Value=:(0.7-8K1P) 01: Current Sensor 02: Meter in Grid 03: Meter in Load 04: 24H consumption 05: PELD OFF 2、Value=:(5-25K) 01: Meter in Grid 02: Meter in Load 03: 24H consumption 04: PELD OFF 3、Value=: (25-50K/50-70K/80-110K) 01: Meter in Grid 02: Meter in Load 03: PELD OFF 04: 24H consumption	
3112	Internal PELD backflow power (PELD Soft Limit for AU 2020 Code)	S16	100W	Value=: 1□□100W + to grid - from grid	Note: Inverter is generating device. Negative is invalid
3113	Internal PELD failsafe switch	U16		Value=: 0: FailSafe off 1: FailSafe on	
3114	Internal PELD real time backflow	U16	10W	Value: 1□□10W	

	power				
3115	Internal PELD Hard Switch for AU 2020 Code	U16		Value=: 00: Null (Not effective) 01: ON 02: OFF	
3116	Internal PELD Hard Limit for AU 2020 Code	S16	100W	Value=: 1□□100W	
3117	Total Energy – Decimal Part	U16	0.001 kW h	Work with 3009-3010 Register	Example :900->0.900kW h
3118-3120	Reserve	U16			
3121	Italy Single Self-Test	U16		Value: Start Single Protection Test 00---Null 01---59.S1(253.0V 3000ms)	Note: The setting has power-off saving function only available under Italy standard.

				02---59.S2(264.5V 200ms) 03---27.S1(195.5V 1500ms) 04---27.S2(34.5V 200ms) 05---81>.S1(50.2Hz 100ms) 06---81<.S1(49.8 Hz 100ms) 07---81>.S2F(51.5Hz 100ms) 08---81<.S2F(47.5 Hz 100ms) 09---81>.S2S(51.5Hz 1000ms) 10---81<.S2S(47.5 Hz 4000ms)	
3122	Italy Full Self-test	U16		Value: Start Complete Self Test 1—In Single self-test condition. 2—In Full Self-test condition	Note: The setting has power-off saving function only available under Italy standard.
3123	01-59.S1 Voltage	U16	0.1V	10□□1V	
3124	01-59.S1 Time	U16	1ms	1□□1ms	

3125	02-59.S2 Voltage	U16	0.1V	10□□1V	
3126	02-59.S2 Time	U16	1ms	1□□1ms	
3127	03-27.S1 Voltage	U16	0.1V	10□□1V	
3128	03-27.S1 Time	U16	1ms	1□□1ms	
3129	04-27.S2 Voltage	U16	0.1V	10□□1V	
3130	04-27.S2 Time	U16	1ms	1□□1ms	
3131	05-81>.S1Frequency	U16	0.01Hz	100□□1Hz	
3132	05-81>.S1 Time	U16	1ms	1□□1ms	
3133	06-81<.S1 Frequency	U16	0.01Hz	100□□1Hz	
3134	06-81<.S1 Time	U16	1ms	1□□1ms	
3135	07-81>.S2F Frequency	U16	0.01Hz	100□□1Hz	
3136	07-81>.S2F Time	U16	1ms	1□□1ms	
3137	08-81<.S2F Frequency	U16	0.01Hz	100□□1Hz	
3138	08-81<.S2F Time	U16	1ms	1□□1ms	
3139	09-81>.S2S Frequency	U16	0.01Hz	100□□1Hz	
3140	09-81>.S2S Time	U16	1ms	1□□1ms	
3141	10-81<.S2S Frequency	U16	0.01Hz	100□□1Hz	
3142	10-81<.S2S Time	U16	1ms	1□□1ms	
3143-3220	Reserved	U16			
3221	Inverter initial setting state	U16		Note: It is mainly the initial setting state returned by DSP. BIT0: Model setting is complete. BIT1: National standard setting is completed. BIT2: The power curve setting is completed. BIT3: Module identification flag (For 110kW models: 0---Onsemi module; 1---Infinoen module). BIT4: Fan detection hardware ID identification flag	

				(For 5-20kW models: 1—supports fan detection, 0—does not support fan detection) BIT5: Australian energy storage machine FCAS function status flag (0, FCAS function is not running, other working modes can be executed; 1, FCAS function is running, does not respond to other working modes) BIT6: AFCI self-test end flag: 1---self-test is over, 0---self-test is not over or self-test is not completed BIT7: AFCI self-test result flag bit: 1---self-test with arc, 0-self-test without arc BIT8, BIT9: Main DSP chip type: 00---F28062,01---F28374S (Analysis: BIT9 is in the front, BIT8 is in the back, 01---BIT9 is 0, BIT8 is 1), this flag is external and Distinguish the role of remote upgrade. BIT10: IGBT screening completion flag bit: 0—screening is not completed or has not been screened, 1—screening is completed. BIT11~BIT13: Resv. BIT14: DSP waveform data flag (0---no waveform; 1---waveform ready). (The default is 0, after the initial setting is successful, it is set to 1, and it can only run after the model and national standard settings are completed)	
3222	Reserved	U16			
3223	Safety Version	U16			
3224-3227	Reserved	U16			
3228	One Click Reset Enable Command Status	U16		0000H---Dsiable 0059H---Enable Use Function Code 06 Address 3302	Note: One Click Reset , Function: Factory Reset and Clear Energy Data
3229	One Click Reset Operating Command Status	U16		0000H---Disable 0045H---Enable Use Function Code 06 Address 3303 If not receive 0045H End flag , HMI will stay in waiting until 5mins timeout	Note: If the operating command is not received within 10s after the enable command is sent, the enable command will automatically become invalid.

3230-3249	SN Number	U16		ASCII Display ie: 3230 = '01' 3231 = '23' Display: '0123'	
3250	Meter placement	U16		BIT00: Meter is on the load side	

				BIT01: Meter is on the main grid side BIT02: Reserved BIT03: Reserved BIT04: Reserved BIT05: Reserved BIT06: Reserved BIT07-BIT15: Reserved	
3251	Meter AC voltage A	U16	0.1V	10<-->1V	
3252	Meter AC current A	U16	0.01A	100<-->1A	
3253	Meter AC voltage B	U16	0.1V	10<-->1V	
3254	Meter AC current B	U16	0.01A	100<-->1A	
3255	Meter AC voltage C	U16	0.1V	10<-->1V	
3256	Meter AC current C	U16	0.01A	100<-->1A	
3257	MeterActive powerA	S32	0.001kW	1000<-->1kW	
3259	MeterActive powerB	S32	0.001kW	1000<-->1kW	
3261	MeterActive powerC	S32	0.001kW	1000<-->1kW	
3263	Meter Total active power	S32	0.001kW	1000<-->1kW	
3265	Meter Reactive power A	S32	1Var	1<-->1Var	
3267	Meter Reactive power B	S32	1Var	1<-->1Var	
3269	Meter Reactive power C	S32	1Var	1<-->1Var	
3271	Meter Total reactive power	S32	1Var	1<-->1Var	
3273	Meter Apparent power A	S32	1VA	1<-->1VA	

3275	Meter Apparent power B	S32	1VA	1<-->1VA	
3277	Meter Apparent power C	S32	1VA	1<-->1VA	
3279	Meter Apparent power	S32	1VA	1<-->1VA	
3281	Meter Power factor	S16	0.001	-1.000~-0.800 +0.800~+1.000 1Ph meter: DDS1352: Actual accuracy 0.001 ACR10RD16TE:Actual accuracy0.01	
3282	Meter Grid frequency	U16	0.01Hz	100<-->1Hz	
3283-3284	MeterGrid power total active energy	U32	0.01kWh	100<-->1kWh	
3285-3286	MeterGrid power transmission total active energy	U32	0.01kW	100<-->1kWh	
3287	PV string voltage and Current Combination Description	U16		Represents several PV voltages corresponding to several PV currents of the string inverter; 0000H: According to 1 voltage and 2 current 0001H: Reserved 0002H: 1 voltage to 2 current 0003H: 1 voltage to 3 current	

				Description: The relationship between PV voltage and current in registers 3289-3335 of 04 function code is associated here. Example: Step1: Get the 3004 register. If the number of PV input channels is greater than 4, continue to the second step; otherwise, directly obtain the DC input voltage and current of the 3022-3029 register address;	
3289	PV21 I	S16	0.1A	10<-->1A	Available for inverters: 1、50-70K Series 2、80-110K(90-136K) Series 3、125K Series 4、250K Series Note: 125K has 20 PV current, 10PV voltage, 1PV voltage -> 2PV
3290	PV22 I	S16	0.1A	10<-->1A	
3291	PV23 I	S16	0.1A	10<-->1A	
3292	PV24 I	S16	0.1A	10<-->1A	
3293	PV25 I	S16	0.1A	10<-->1A	
3294	PV26 I	S16	0.1A	10<-->1A	
3295	PV27 I	S16	0.1A	10<-->1A	

3296	PV28 I	S16	0.1A	10<-->1A	current 230K series 30 strings
3297	PV29 I	S16	0.1A	10<-->1A	
3298	PV30 I	S16	0.1A	10<-->1A	
3299	Total PV V	U16	0.1V	10<-->1V	
3300	Total PV I	S16	0.1A	10<-->1A	
3301	PV1 I	S16	0.1A	10<-->1A	
3302	PV2 I	S16	0.1A	10<-->1A	
3303	PV3 I	S16	0.1A	10<-->1A	
3304	PV4 I	S16	0.1A	10<-->1A	
3305	PV5 I	S16	0.1A	10<-->1A	
3306	PV6 I	S16	0.1A	10<-->1A	
3307	PV7 I	S16	0.1A	10<-->1A	
3308	PV8 I	S16	0.1A	10<-->1A	
3309	PV9 I	S16	0.1A	10<-->1A	
3310	PV10 I	S16	0.1A	10<-->1A	
3311	PV11 I	S16	0.1A	10<-->1A	
3312	PV12 I	S16	0.1A	10<-->1A	
3313	PV13 I	S16	0.1A	10<-->1A	
3314	PV14 I	S16	0.1A	10<-->1A	
3315	PV15 I	S16	0.1A	10<-->1A	
3316	PV16 I	S16	0.1A	10<-->1A	
3317	PV17 I	S16	0.1A	10<-->1A	
3318	PV18 I	S16	0.1A	10<-->1A	
3319	PV19 I	S16	0.1A	10<-->1A	
3320	PV20 I	S16	0.1A	10<-->1A	
3321	PV1 V	U16	0.1V	10<-->1V	
3322	PV2 V	U16	0.1V	10<-->1V	
3323	PV3 V	U16	0.1V	10<-->1V	
3324	PV4 V	U16	0.1V	10<-->1V	

3325	PV5 V	U16	0.1V	10<-->1V	
3326	PV6 V	U16	0.1V	10<-->1V	
3327	PV7 V	U16	0.1V	10<-->1V	
3328	PV8 V	U16	0.1V	10<-->1V	
3329	PV9 V	U16	0.1V	10<-->1V	
3330	PV10 V	U16	0.1V	10<-->1V	
3331	PV11 V	U16	0.1V	10<-->1V	
3332	PV12 V	U16	0.1V	10<-->1V	
3333	PV13 V	U16	0.1V	10<-->1V	
3334	PV14 V	U16	0.1V	10<-->1V	
3335	PV15 V	U16	0.1V	10<-->1V	
3336-3340	Reserve	U16			
3341	This model IV curve number	U16		0--- 0 IV curve; 1--- 1 IV curve, 2--- 2 Max 30	
3342	Current IV curve No.	U16		0--- 0 IV curve; 1--- 1 IV curve, 2--- 2....., Max 30	
3343	PVV1	U16	0.1V	10<-->1V	
3344	PVI1	S16	0.1A	10<-->1A	
3345	PVV2	U16	0.1V	10<-->1V	
3346	PVI2	S16	0.1A	10<-->1A	
3347	PVV3	U16	0.1V	10<-->1V	
3348	PVI3	S16	0.1A	10<-->1A	
3349	PVV4	U16	0.1V	10<-->1V	
3350	PVI4	S16	0.1A	10<-->1A	

3351	PVV5	U16	0.1V	10<-->1V	
3352	PVI5	S16	0.1A	10<-->1A	
3353-3460		...	...		
3461	PVV60	U16	0.1V	10<-->1V	
3462	PVI60	S16	0.1A	10<-->1A	
3463-3499	Reserve	U16		35	
3500	MPPT 1V	U16	0.1V	10<-->1V	Available for : 1、50-70K Series 2、80-110K (90-136K) Series 3、125K Series 4、250K Series
3501	MPPT 2V	U16	0.1V	10<-->1V	
3502	MPPT 3V	U16	0.1V	10<-->1V	
3503	MPPT 4V	U16	0.1V	10<-->1V	
3504	MPPT 5V	U16	0.1V	10<-->1V	
3505	MPPT 6V	U16	0.1V	10<-->1V	

3506	MPPT 7V	U16	0.1V	10<-->1V	
3507	MPPT 8V	U16	0.1V	10<-->1V	
3508	MPPT 9V	U16	0.1V	10<-->1V	
3509	MPPT 10V	U16	0.1V	10<-->1V	
3510	MPPT 11V	U16	0.1V	10<-->1V	
3511	MPPT 12V	U16	0.1V	10<-->1V	
3512	MPPT 13V	U16	0.1V	10<-->1V	
3513	MPPT 14V	U16	0.1V	10<-->1V	
3514	MPPT 15V	U16	0.1V	10<-->1V	
3515-3529	Reserve	U16			
3530	MPPT 1I	S16	0.1A	10<-->1A	Available for : 1、50-70K Series 2、80-110K (90-136K) Series 3、125K Series
3531	MPPT 2I	S16	0.1A	10<-->1A	
3532	MPPT 3I	S16	0.1A	10<-->1A	

3533	MPPT 4I	S16	0.1A	10<-->1A	4、250K Series
3534	MPPT 5I	S16	0.1A	10<-->1A	
3535	MPPT 6I	S16	0.1A	10<-->1A	
3536	MPPT 7I	S16	0.1A	10<-->1A	
3537	MPPT 8I	S16	0.1A	10<-->1A	
3538	MPPT 9I	S16	0.1A	10<-->1A	
3539	MPPT 10I	S16	0.1A	10<-->1A	
3540	MPPT 11I	S16	0.1A	10<-->1A	
3541	MPPT 12I	S16	0.1A	10<-->1A	
3542	MPPT 13I	S16	0.1A	10<-->1A	
3543	MPPT 14I	S16	0.1A	10<-->1A	
3544	MPPT 15I	S16	0.1A	10<-->1A	
3545- 3599	Reserve	U16			
3700	Rated active power	U32	W	1000-1KW 3700 is high, 3701 is low	Applicable to all US versions.
3701	Rated active power	U32	W		
3702	Active power at PF=+0.8 under rated full load	U32	W	1000-1KW High	
3703	Active power at PF=+0.8 under rated full load	U32	W	Low	
3704	Leading power factor +0.8	U16	0.01	1->0.01	
3705	Active power at PF=-0.8 under rated full load	U32	W	1000-1KW High	
3706	Active power at PF=-0.8 under rated full load	U32	W	Low	
3707	Lagging power factor -0.8	U16	0.01	-1 -> -0.01	

3708	Normal operating	U16		Other-Null	
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	performance category			1-CAT_A 2-CAT_B	
3709	Abnormal operating performance category	U16		Other-Null 1- CAT_I 2- CAT_II 3- CAT_III	
3710	AC rated voltage	U16	V	480->480V	
3711	AC 1st overvoltage upper limit	U16	0.1V	3326->332.6V AC 1st overvoltage upper limit	
3712	AC 1st undervoltage lower limit	U16	0.1V	138->13.8V AC 1st undervoltage lower limit	
3713	Control mode	U16	V	Bit0:Frequency derating Bit1:Volt-watt Bit2:Volt-var Bit3:Fixed Reactive Power Bit4:Fixed PF Bit5:P-Q	
3714	Admittance parameters	U16	0.001 0.0000 0 001	1000->1---80-110K Series 10000000->1-10K Description: 1-10K (excluding 6-10K) series single phase	
3715	Manufacturer	U16		Null Autarco	
3716	Hardware version number (mainboard)	U16		Null 0x1010:DK1635-V2(1ph 1-10K, Exclude 6-10K) 0x1020:IVDG2330-V4 (80-100K) 0x1021:IVDDF2330-V5.2 (230K)	
3717~ 3 718	Real-time reactive power	U32		1000-1KW	

3719-3720	Capacitive maximum reactive power, 60% apparent power	U32	Var		
3721-3722	Inductive maximum reactive power, 60% apparent power	U32	Var		
3723-3830	Reserved	U16			

5.3. Register address of inverter setting

The function code is 0x03, 0x06 and 0X10, the register address needs to offset one bit.

Example: register address: 3000, the send address is 2999.

Register address	name	Data type	Unit	Remark	Note
3000	Year	U16		00-99	
3001	Month	U16			
3002	Day	U16			
3003	Hours	U16			
3004	Mins	U16			
3005	Seconds	U16			
3006	Slave address	U16		The slave address setting is only set with the 0x06 function code;	
3007	ON/OFF	U16		0xBE-ON 0xDE-OFF 10H—Night ON enable 11H—Night ON disable 20H—24H Consumption Enable 21H—24H Consumption Disable	
3008	Reserve				

3009	Power curve No.	U16			
3010	81>S1	U16	0.01Hz	100<-->1Hz, 48-53Hz, default 50.2Hz 43038-43049 Only valid under CEI 0-21 standard	
3011	81>S1-T	U16	0.01s	100<-->1S, 0.05-0.5S, Default :0.10S	
3012	81<S1	U16	0.01Hz	100<-->1Hz, 48-53Hz, Default :49.8Hz	
3013	81<S1-T	U16	0.01s	100<-->1S, 0.05-0.5S, Default :0.10S	
3014	81>S2F	U16	0.01Hz	100<-->1Hz, 48-53Hz, Default :51.5Hz	
3015	81>S2F-T	U16	0.01s	100<-->1S, 0.05-0.5S, Default :0.10S	
3016	81<S2F	U16	0.01Hz	100<-->1Hz, 45-49Hz, Default :47.5Hz	
3017	81<S2F-T	U16	0.01s	100<-->1S, 0.05-0.5S, Default :0.10S	
3018	81>S2S	U16	0.01Hz	100<-->1Hz,	

				48-53Hz, Default :51.5Hz	
3019	81>S2S-T	U16	0.01s	100<-->1S, 0.05-5.0S, Default :1.0S	
3020	81<S2S	U16	0.01Hz	100<-->1Hz, 45-49Hz, Default :47.5Hz	
3021	81<S2S-T	U16	0.01s	100<-->1S, 0.05-5.0S, Default :4.0S	
3022	Reserve	U16			
3023	DRM S1 Limit 1	U16		10000<-->100% Range (0-100%)	
3024	DRM S1 Limit 2	U16		10000<-->100% Range (0-100%)	

3025	DRM S1 Limit 3	U16		10000<-->100% Range (0-100%)	
3026	DRM S1 Limit 4	U16		10000<-->100% Range (0-100%)	
3027	DRM ON/OFF	U16		0x0000 — OFF 0x00AA — ON Default: 0。	Note: After this is turned on, the PELD function will be turned off automatically.
3028	Restart the HMI	U16		0xAA55 is valid, others are invalid Only for single register write, it will take effect if it is successfully sent three times within 6 seconds.	
3029	Inverter installation Method			Inverter installation method (default 0; 0 is 90° installation method, 1 is 15° installation method; according to different installation methods, the inverter implements different temperature derating logic)	
3030	Night SVG Q Set	S16		10000<-- >100%; Range : -60%--- 60%; Default :0%	
3031-3050	Reserve	U16			
3051	Reactive power limitation	S16	1%	10000<-- >100% Range (-6000- +6000) default: 0 Only available for working mode 04	Note: To enable 3051, need to enable 3071 A1. To have power off saving function, need to work with

					3069
3052	Power limitation	U16	1%	10000<-- >100% Range (0-110%) 100% = rated.	Note: To enable 3052, need to enable 3070 AA. To have power off saving function, need to work with 3069
3053	PF Setting	S16	0.001	PF: (800<-- >0.80 , 1000<-- >1.00) (-800<-- >-0.80 , -1000<-- >-1.00) (PF 1.00 same with -1.00) range (-1~-0.80---0.80~1)	Note: To have power off saving function, need to work with 3069
3054	PF Setting 02	S16	0.001	PF Low: (800<-- >0.80 , 1000<-- >1.00) (-800<-- >-0.80 , -1000<-- >-1.00) (PF 1.00 same with -1.00) range (-1~-0.80---0.80~1) For working mode 03 fixed pf function	Note: To enable 3054, need to enable 3071 A2. To have power off saving function, need to work with 3069
3055	Reserve				
3056- 3057	Calibrate total generation	U32	1kWh		
3058- 3059	Calibrate current month generation	U32	1kWh		
3060- 3061	Calibrate last month generation	U32	1kWh		
3062	Calibrate today generation	U16	0.1kW h		
3063	Calibrate yesterday generation	U16	0.1kW h		

3064-3065	Calibrate this year generation	U32	1kWh		
3066-3067	Calibrate last year generation	U32	1kWh		
3068	Standard number	U16		See Appendix 2	
3069	Power-off saving function Note: Don't set 1 too frequently, the flash has a limited write and read	U16		BIT00: 0- Power off not saving 1- Power off saving	For 3052/3149/3150 Reg
				BIT01:	For 3053 Reg

	lifespan. Less than 10000 times.			0- Power off not saving 1- Power off saving	
				BIT02: 0- Power off not saving 1- Power off saving	For 3051, 3130-3146,3073, 3126-3127,3321-3332Reg
				BIT03: 0- Power off not saving 1- Power off saving	For 3051, 3130-3146,3073, 3126-3127,3321-3332Reg
				BIT04-15:	Reserved
3070	Power limitation switch	U16		0xAA ON, 0x55 OFF(Power to 100%)(for 3052 and 3081 Reg)。	default :0xAA。
3071	Reactive power switch	U16		0x55 OFF, PF=1 Reactive =0 0xA1 Reactive setting effective, PF =1 (for 3051 and 3083 Reg); 0xA2 PF 02 setting effective, Reactive =0 (for 3054 Reg) 。	default :0x55。

3072	Day Time PID Switch	U16		0xAA ON 0x55 OFF Default: OFF	Note: Available for 255K Series DO set with Day Time PID Solution
3073	Working mode	U16		Working mode: 00---No response mode 01---Volt-watt default 02---Volt-var 03---Fixed power factor 04---Fix reactive power 05---Power-PF 06---Rule21 Volt-watt 0C--- P-Q	Note: If need power off saving function, set 3069=1
3074	Italy Standard Switching Mode	U16		0: Off, Default: OFF (81>S1—50.2Hz/0.1s-Inverter 81<S1—49.8Hz/0.1s-Inverter) 1: Local Control (81>S2—51.5Hz/0.1s-Local 81<S2—47.5Hz/0.1s-Local) 2: External Signal (81>S2—51.5Hz/1.0s-External 81<S2—47.5Hz/4.0s-External)	Note: The setting has power-off saving function only available under Italy standard.

3075	Italy Single Test	U16		Value: Start Single Protection Test 00---Null 01---59.S1(253.0V 3000ms) 02---59.S2(264.5V 200ms) 03---27.S1(195.5V 1500ms) 04---27.S2(34.5V 200ms) 05---81>.S1(50.2Hz 100ms) 06---81<.S1(49.8 Hz 100ms) 07---81>.S2F(51.5Hz 100ms) 08---81<.S2F(47.5 Hz 100ms) 09---81>.S2S(51.5Hz 1000ms) 10---81<.S2S(47.5Hz 4000ms)	Note: The setting has power-off saving function only available under Italy standard.
3076	Italy Full Test	U16		Value: Start Complete Self Test 0---Full test stop or not start 2---Full test starts	Note: The setting has power-off saving function only available under Italy standard.
3077	AFCI ON/OFF	U16		1 <--> ON 0 <--> OFF	
3078	CT Direction Set	U16		0xAA Correct, 0x55 Reverse, Others not effective. Correct direction is towards grid. When actual connection is reversed, set 0x55 can work normally.	
3079	Reserve	U16			

3080	Power control word	U16		iPower control word: Bit0---Max power limit flag: 0---Default is 1.09 rated P (For 4777-A/B/C/N, UL,Mex code, receive 0, indicates 1.00 rated P; Others default 1.09 rated P) 1---Set as 1.1 rated P When HMI or external 485 set once,ARM will save this flag in the flash and detect it after power on and send DSP the command Bit1---Resv Bit2— Remote active power limit enable flag bit (associated with 3031 remote control active power percentage register address) 1---enable; 0---dsable; default is 0
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				(Note: it needs to be saved after power-off, and when this bit is 1, it needs to be sent after power-on)	
3081	Limit power actual value	S16	10W	1<-- >10W Range:-327680W~327680W Use 06 code to open 3070 Reg, Then set 3081 Reg	
3082	Reserve	U16			
3083	Limiting reactive power adjustment value	S16	10Var	1<-- >10Var Range: -327680Var~327680Var Use 06 code to open 3071 Reg, Then set 3083 Reg	
3084	Leakage current protection	U16	1mA	1<-- >1mA Range : 50-800mA ; Default 240mA	
3085	PV insulation protection	U16	1kOhm	1<-- >1k Range : 20k-1000k;	Default 200K
3086-3088	Reserve	U16			

3089	Grid code accuracy set flag	U16	The lower 8 bits represent the modifiable precision setting parameters: 0x0001: Set the precision selection voltage 0.1V, time precision 0.01s, frequency precision 0.01Hz. 0x0000: Set the precision selection voltage 1V, the time precision is 0.1s, and the frequency precision is 0.1Hz. The upper 8 bits represent unmodifiable setting parameters: 0x01xx: a single rule, the setting precision selects the voltage 0.1V, the time precision is 0.02s, and the frequency precision is 0.01Hz. Note: 1. The upper computer first reads 3089 "Grid code accuracy set flag" through the 03 function code, and then performs corresponding settings according to the precision requirements. If the value of 3089 is less than or equal to 255, you can read and set the precision of the relevant national standard by modifying the value. This value will not be saved when power off. If the value of 3089 is greater than 255, the value cannot be modified. 2. The 0x01xx mode is an added function according
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			to the IEEE1547 standard (UL-208V-A2 and UL-240V-A2) when the over-under-frequency time exceeds 1000s. 3. The register address range covered by 3089 is: 3090-3105.	
3090	OV-G-V 01	U16	See 3089 register address definition rules for numerical precision User-def Code Range 1P range: 240-270V default: 260V 3P range: 236-335V default: 254V	Note: User-defined standard setting must use function code 0x10

3091	OV-G-V-T 01	U16		See 3089 register address definition rules for numerical precision User-def Code Range Range: 0.10-9.0s default: 1.0s	
3092	OV-G-V 02	U16		See 3089 register address definition rules for numerical precision User-def Code Range 1P range: 240-300V default: 280V 3P range: 248-341V default: 265V	
3093	OV-G-V-T 02	U16		See 3089 register address definition rules for numerical precision User-def Code Range Range: 0.10-1.0s default: 0.2s	
3094	UN-G-V 01	U16		See 3089 register address definition rules for numerical precision User-def Code Range 1P range: 170-210V default: 180V 3P range: 173-236V default: 190V	
3095	UN-G-V-T 01	U16		See 3089 register address definition rules for numerical precision User-def Code Range	

				Range: 0.10-9.0s default: 1.0s	
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3096	UN-G-V 02	U16		See 3089 register address definition rules for numerical precision User-def Code Range 1P range: 110-210V default: 180V 3P range: 132-219V default: 173V	
3097	UN-G-V-T 02	U16		See 3089 register address definition rules for numerical precision User-def Code Range Range: 0.10-1.0s default: 0.2s	
3098	OV-G-F 01	U16		See 3089 register address definition rules for numerical precision User-def Code Range Range: 50.2-53.0Hz default: 51.0Hz Range: 60.2-63.0Hz default: 61.0Hz	
3099	OV-G-F-T 01	U16		See 3089 register address definition rules for numerical precision User-def Code Range Range: 0.10-9.0s default: 1.0s	
3100	OV-G-F 02	U16		See 3089 register address definition rules for numerical precision User-def Code Range Range: 51.0-53.0Hz default: 51.0Hz Range: 61.0-63.0Hz default: 51.0Hz	
3101	OV-G-F-T 02	U16		See 3089 register address definition rules for numerical precision User-def Code Range Range: 0.10-9.0s default: 0.2s	

3102	UN-G-F 01	U16		See 3089 register address definition rules for numerical precision User-def Code Range	
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				Range : 47.0-49.5Hz default: 48.0Hz Range : 57.0-59.5Hz default: 48.0Hz	
3103	UN-G-F-T 01	U16		See 3089 register address definition rules for numerical precision User-def Code Range Range : 0.10-9.0s default: 1.0s	
3104	UN-G-F 02	U16		See 3089 register address definition rules for numerical precision User-def Code Range Range : 47.0-49.0Hz default: 47.0Hz Range : 57.0-59.0Hz default: 47.0Hz	
3105	UN-G-F-T 02	U16		See 3089 register address definition rules for numerical precision User-def Code Range Range : 0.10-9.0s default: 0.2s	
3106	Startup time	U16	1s	1<-->1s Range : 10-600s default: 60s	
3107	Reconnect time	U16	1s	1<-->1s Range : 10-600s default: 60s	
3108	Recover V upper limit	U16	0.1V	10<-->1V ; Resolution 0.1V ; Default is the OVG01	
3109	Recover V lower limit	U16	0.1V	10<-->1V ; Resolution 0.1V ; Default is the UNG01	

3110	Recover F upper limit	U16	0.01Hz	100<-->1Hz; Resolution 0.1Hz; Default is the OVGf01	
3111	Recover F lower limit	U16	0.01Hz	100<-->1Hz; Resolution 0.1Hz; Default is the UNGf01	
3112	Startup V upper limit	U16	0.1V	10<-->1V; Resolution 0.1V; Default is 253.0V	
3113	Startup V lower limit	U16	0.1V	10<-->1V; Resolution 0.1V; Default is 195.5V	
3114	Startup F upper limit	U16	0.01Hz	100<-->1Hz; Resolution 0.1Hz; Default is 50.1Hz	
3115	Startup F lower limit	U16	0.01Hz	100<-->1Hz; Resolution 0.1Hz; Default is 49.5Hz	
3116	Denmark Freq Change Protection Threshold	U16	0.1Hz/s	(1000<-->1Hz/s, Range:0-3500, Default:2500)	

				Note: LCD display Hz/s, range 0-3.5Hz/s, step 0.1Hz/s	
3117	Denmark Freq Change Protection Time	U16	1ms	(1<-->1ms), Range 0-5000, Default 80	
3118	Working mode control switch	U16		BIT00, overvoltage active power automatic limit, 0 off (default), 1 on BIT01, Vref control enable (3126-3127 register address), 0 off (default), 1 on BIT02-15, Reserved	Note: For Italian and Polish PN50549 standard requirements.
3119	OV-G-V03	U16	0.1V	1<-->0.1V	
3120	OV-G-V03-T	U16	10ms	1<-->10mS	

3121	UN-G-V03	U16	0.1V	1<-->0.1V R14-208 Range : 41.6-183.0V Default : 104.0V R14-240 Range : 48.0-211.2V Default : 120.0V	
3122	UN-G-V03-T	U16	10ms	1<-->10mS Range : 0.5-21.0s Default : 2.0s	
3123- 3125	Reserve	U16			
3126	Volt-Var Vref Setting	U16	0.1V	10<-->100% Range : 263.1~290.9V Default: 277V Note: The power-off save function needs to be enabled in the 3069 register BIT03 or BIT02. Set Vref first, then set the voltage value of mode 2 (register address 3138-3141)	Note: The Vref setting function needs to open the BIT01 Vref control enable bit of the 3118 register;
3127	Volt-Var Vref Time Setting	U16		5000<-->500s Range : 300s~5000s Default: 300s Note: The power-off save function needs to be enabled in the 3069 register BIT03 or BIT02.	Note: The Vref setting function needs to open the BIT01 Vref control enable bit of the 3118 register;
3128	02 working mode V2-Q2 Percent	S16	1%	10000<-- >100% Range : -60%--- 60%; Default: 00% >0 is leading; <0 is lagging	

3129	02 working mode V3-Q3 Percent	S16	1%	10000<-- >100% Range : -60%--- 60%; Default: 00% >0 is leading;	
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				<0 is lagging	
3130	01/06 working mode V1Set	U16	0.1V	10<-->1V; If need power off saving, set 3069 BIT03 or 02	
3131	01/06 working mode V2Set	U16	0.1V	10<-->1V;	Power-off save function, need to open 3069 register BIT03 or BIT02
3132	01/06 working mode V3Set	U16	0.1V	10<-->1V;	
3133	01/06 working mode V4Set	U16	0.1V	10<-->1V;	
3134	01/06 working mode(P1% Set)	U16	1%	10000<-->100% Range(0-100%) 100%- P-rated	
3135	01/06 working mode(P2% Set)	U16	1%	10000<-->100% Range(0-100%) 100%- P-rated	
3136	01/06 working mode(P3% Set)	U16	1%	10000<-->100% Range(0-100%) 100%- P-rated	
3137	01/06 working mode(P4% Set)	U16	1%	10000<-->100% Range(0-100%) 100%- P-rated	
3138	02 working mode V1Set	U16	0.1V	10<-->1V; If need power off saving, set 3069 BIT03 or 02 First set Vref (3126-3127), then set the voltage value of mode 2	
3139	02 working mode V2Set	U16	0.1V	10<-->1V; If need power off saving, set 3069 BIT03 or 02 First set Vref (3126-3127), then set the voltage value of mode 2	
3140	02 working mode V3Set	U16	0.1V	10<-->1V; If need power off saving, set 3069 BIT03 or 02 First set Vref (3126-3127), then set the voltage value of mode 2	

3141	02 working mode V4Set	U16	0.1V	10<-->1V; First set Vref (3126-3127), then set the voltage value of mode 2	
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3142	02 working mode (MaxLeadingVar%)	U16	1%	10000<-->100% Range: 0--- 60%; Default: 30%	If need power off saving, set 3069 BIT03 or 02
3143	02 working mode (MaxLaggingVar%)	U16	1%	10000<-->100% Range: 0--- (-60%); Default: -30%	
3144	05 working mode (Pb% Set)	U16	1%	(10000<-->100%); Range: 50--- 100%; Default: 100%	
3145	05 working mode (Pc% Set)	U16	1%	(10000<-->100%); Range: 50---100%; Default: 100%	
3146	05 working mode (PFcSet)	S16	0.001	Range: (-0.90--- 0.90); Default: -0.95 (800<-->0.80, 1000<-->1.00)	
3147	10mins over voltage	U16	0.1V	10<-->1V;	
3148	Power ramp rate (Wgra), general	U16	1%	: (10000<-->100%); Range: 5%---600%; Default: 16.67%; Accuracy 1% Start up ramp rate	
3149	Power ramp up rate	U16	1%	3000<-->30%/min; Range: 10%—100%; Default: 16.66%, Only for AUS If need power off saving, set 3069 as 1	
3150	Power ramp down rate	U16	1%	3000<-->30%/min; Range: 10%—100%; Default: 50% , Only for AUS If need power off saving, set 3069 as 1	

3151	Internal PELD Switch (PELD Soft Switch for AU 2020 Code)	U16		1、Value= : (0.7-8K1P) 01 : Current Sensor 02 : Meter in Grid 03 : Meter in Load 04 : 24H consumption 05: PELD OFF 2、Value= : (3-20K 3P) 01 : Meter in Grid 02 : Meter in Load 03 : 24H consumption 04 : PELD OFF 3、Value= : (25-50K/50-70K/80-110K) 01 : Meter in Grid 02 : Meter in Load	
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				03 : PELD OFF 04 : 24H consumption	
3152	Internal PELD backflow power (PELD Soft Limit for AU 2020 Code)	S16	100W	Value=: 1□□100W + to grid - from grid	
3153	Internal PELD failsafe switch	U16		Value=: 0: FailSafe off 1: FailSafe on	
3154	Internal PELD Hard Switch for AU 2020 Code	U16		Value=: 00: Null (Not effective) 01 : ON (For Meter in Grid) 02 : OFF 03: ON (For Meter in Load) For all AU 2020 Code Series	Note: If soft switch is ON, only 01 and 02 is effective If soft switch is OFF, 01-03 are effective, for individual hard switch
3155	Internal PELD Hard Limit for AU 2020 Code	S16	100W	Value=: 1□□100W + to grid - from grid	

3156-3239	Reserve	U16			
3240	IV curve starting setting	U16		Start, Don't scan, Default is 0 Note: After start the IV curve scanning, use function code 04, read 3341Reg. If READ THE IV curve number, the scan ends. If no number after 5mins, end with timeout. If 3 times of timeout, scanning error	Support Model 25-50k 50-70k 80-110(90-136) k 125k、230k
3241	IV curve start voltage	U16	1V	1<-->1V	
3242	IV curve interval voltage	U16	1V	1<-->1V	
3243	Get current IV curve number	U16		Note: Max number is 30 Step 1: Send IV curve scanning command, wait it ends Step 2: Send function 06 3243 Reg to set IV curve number, get IV curve info	

				Step 3: Use function 04 read IV curve info Reg from 3341-3462	
3244-3300	Reserve	U16			
3301	Fan self-test	U16		0000H—Fan self test off 0001H—Fan self test On After send self-test command, check 04 function code 3044 register. If detect F011H, it means fan is abnormal	

3302	One Click Reset Enable Command	U16		0000H---Disable 0001H---Enable, Use Function Code 04 Address 3328 Get random value xxxxH for enable	Note : One Click Reset, Function: Factory Reset and Clear Energy Data
3303	One Click Reset Operating Command	U16		0000H---Disable xxxxH---Enable, 1、xxxxH Enable Value is from function code 04 address 3328 2、Read the 3229 register address through the 04 function code, and get 0045H ('E'), which means the end of one click reset. Or force a wait of 5 minutes and the timeout expires.	Note: If the operating command is not received within 10s after the enable command is sent, the enable command will automatically become invalid.
3304	Special Function Control Word	U16		Special Function control word BIT0-BIT15 (Power off saving) BIT00 : Boost not working function OFF Control word 0---ON; 1---OFF; Default 0 Note: OFF-> Boost always working BIT01 : DC injection adjustment function off control word 0---ON; 1---OFF; Default 0 Note: OFF-> Stop DC injection adjustment, for test purpose BIT02 : 0% Power Relay Trip Function Switch 1---Enabled; 0--Disabled; Default is 0 (Note: When power control is set to 0%, the AC relay will	

				disconnect if this function is enabled) BIT03-BIT04 : Reserved BIT05 : L-Ground Fault Detection Function Switch; 0---Enabled; 1---Disabled; Default:0 BIT06 : Reserved BIT07 : AC hardware overcurrent low sensitivity switch flag bit; 0---low sensitivity protection function is off (2 to 5 protections within 30 seconds), 1---low sensitivity protection function is turned on (50 protections within 30 seconds, equivalent to shielding The hardware overcurrent protection is disabled, but the blocking wave logic operates normally). ; default is 0 BIT08-15 : Reserved	
3305	Grid V A compensation	S16	0.1V	10<-->1V Range: -5.0V~+5.0V Default 0 1Ph inverter only phase A	
3306	Grid V B compensation	S16	0.1V	10<-->1V Range: -5.0V~+5.0V Default 0	
3307	Grid V C compensation	S16	0.1V	10<-->1V Range: -5.0V~+5.0V Default 0	
3308-3310	Reserved				
3311	Enable “Power ON display to select grid code” Flag	U16		00: Disable (Power ON NOT display to select grid code) 01: Enable (Power ON display to select grid code) 00 is default	

3312	Special control word 02	U16		Special function control word 02- BIT0-BIT15 (the control word is saved after power off) BIT00: constant voltage Mppt mode enable control bit; (Note: the associated 3313 register is used at the same time) 0---disable; 1---enable; Default is 0; BIT01: Multi-channel Mppt parallel enable control bit; 0--- Mppt runs independently for each channel; 1--- Mppt runs in parallel;
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				The default is 0; BIT02-15: Reserved.	
3313	Voltage value in constant voltage Mppt mode	U16	0.1V	10<-->1V Range 100V---850V;	Description: The associated 3312 registers are used at the same time
3314-3319	Reserved	U16			
3320	US Code reconnection power recovery slope setting value	U16	0.01%	1->0.01% Range: 0.1%~100%, Default: 0.33%	
3321	P_Q Mode P1	U16	100	100->1% Range: 0~70%Pn, Default : 20%Pn	Note: The power-off save function needs to be enabled in the 3069 register BIT03 or BIT02.
3322	P_Q Mode P2	U16	100	100->1% Range: 40%~80%Pn, Default: 50%Pn	
3323	P_Q Mode P3	U16	100	100->1% Range: 50%~100%Pn, Default: 100%Pn	
3324	P_Q Mode P4	U16	100	100->1%	
3325	P_Q Mode P5	U16	100	100->1%	

3326	P_Q Mode Q1	U16	100	100->1% Range: -60~60%Sn, Default: 0Sn	
3327	P_Q Mode Q2	U16	100	100->1% Range: -60%~60%Sn, Default: 0Sn	
3328	P_Q Mode Q3	U16	100	100->1% Range: -60%~60%Sn, Default: -44%Sn	
3329	P_Q Mode Q4	U16	100	100->1%	
3330	P_Q Mode Q5	U16	100	100->1%	
3331	Volt Var P3Tau		0.01s	1->0.01s Range: 0.5~60s, Default: 10s	
3332	Volt Var Q3Tau		0.01s	1->0.01s Range: 1~90s, Default: 5s	
3333-3400	Reserved				
3401	UnderFrequency Ramping Fstart	U16	0.01Hz	1->0.01Hz Range: 59Hz~59.8Hz, Default: 59.8Hz	
3402	UnderFrequency Ramping	U16	0.01	1->0.01%	

	Droop			Range: 2%~5%, Default: 5%	
3403	Frquency Derating Response Time	U16	0.01s	1->0.01s Range: 0.2s~10s, Default: 5s	
3404	Frquency Derating Droop	U16	0.01	1->0.01% Range: 2%~5%, Default: 5%	
3405	Frquency Derating Fstart	U16	0.01Hz	1->0.01Hz Range: 60.01Hz~61Hz, Default: 60.2Hz	

3406-3500	Reserve				
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5.4. PELD (external device) operating information

Function code 0x04. No need address offset

Note: The following information applies to external PELD devices

Register address	name	Data type	Unit	Remark	Address type
36000	PELD AC V A	U16		10<-->1V	3X
36001	PELD AC I A	U16		10<-->1A	3X
36002	PELD AC V B	U16		10<-->1V	3X
36003	PELD AC I B	U16		10<-->1A	3X
36004	PELD AC V C	U16		10<-->1V	3X
36005	PELD AC I C	U16		10<-->1A	3X
36006	PELD_Power A	S16		1<-->100W	3X
36007	PELD_Power B	S16		1<-->100W	3X
36008	PELD_Power C	S16		1<-->100W	3X
36009-36010	PELD_Power Total	S32		1<-->100W Note: Little Endian Low first, High Latter	3X
36011-36012	Inverter Total Power	S32		1<-->100W Note: Little Endian Low first, High Latter	3X
36013-36014	Reserve				3X
36015	Power control percent	U16		10000<-->100%	3X
36016	CT ratio	U16		1<-->10 Note:PELD-MI/MII, 05+ Ver change to 1-10, Original is 1-100	3X
36017	Backflow power setting value	S16		1<-->100W	3X

36018	Inverter number setting value	U16			3X
36019	Year	U16		00-99	3X
36020	Month	U16			3X
36021	Day	U16			3X
36022	Hours	U16			3X
36023	Mins	U16			3X
36024	Seconds	U16			3X
36025	FailSafe ON/OFF	U16		0□□OFF 1□□ON Default:OFF, After Ver06	3X
36026	Grid PF	S16		-1.0~-0.8 +0.8~+1.0	3X
36027	Grid Freq(Meter)	U16	0.01Hz	100<-->1Hz	3X
36028-3 602 9	Total Load power	U32		1<-->100W Note: Calculate from inverter power and PELD power Note: Little Endian Low first, High Latter	3X
36030-3 604 9	Reserve				3X
36050-3 605 1	Inverter total generation energy	U32	0.01kWh	100<-- >1kWh Note: Little Endian Low first, High Latter	3X
36052-3 605 3	Load total consumption energy	U32	0.01kWh	100<-- >1kWh Note: Little Endian Low first, High Latter	3X
36054-3 605 5	Grid import total active energy	U32	0.01kWh	100<-- >1kWh Note: Little Endian Low first, High Latter	3X
36056-3 605 7	Grid export total active energy	U32	0.01kWh	100<-- >1kWh Note: Little Endian Low first, High Latter	3X

36058	PELD data transmission Switch (Transmit ON/OFF)	U16		0□□OFF; 1□□ON; Default is 0;	3X
36059	Batch upgrade flag	U16		0□□ Not support batch upgrade; 1□□ support batch upgrade; Default is 0;	3X
36060	PELD model	U16		00E0: PELD-MI/MII 0000: Unknown PELD (Hex)	3X

5.5. PELD (External device) setting

Function code 0x03,0x06 and 0x10 No need off set.

Register address	name	Data type	Unit	Remark	Address type
36500	Year	U16		00-99	4X
36501	Month	U16			4X
36502	Day	U16			4X
36503	Hours	U16			4X
36504	Mins	U16			4X
36505	Seconds	U16			4X
36506	Slave Address	U16			4X
36507	CT ratio	U16		1<-->100	4X
36508	Backflow power	S16		1<-->100W	4X
36509	Inverter number	U16			4X
36510	PELD mode	U16		01 : Sum of three phase power 02: Minimum power from one phase *3。	4X

36511	FailSafe ON/OFF	U16		0□□OFF 1□□ON Default:OFF	4X
36512	Reserve				4X
36513	PELD data transmission Switch	U16		0□□OFF; 1□□ON; Default is 0;	4X
36514	Inverter type Select	U16		0x55□□PV inverter(Meter Info OFF) ; 0xAA□□Hybrid inverter (Meter Info ON) ; Default: 0x55;	4X

6. Example

6.1. 04 function code

1) Acquire 1 operating message

Send		Response	
01 04 0B B7 00 01 83 C8		01 04 02 00 43 F8 C1	
01	Address	01	Address
04	Function code	04	Function code
0B B7	Register Start Addr	0B B7	Register Start Addr
00 01	1 register	00 01	1 register
83 C8	CRC Check	83 C8	CRC Check

2) Acquire multiple operating message

Send		Response	
01 04 0B B7 00 02 C3 C9		01 04 04 00 43 02 07 4A F2	
01	Address	01	Address
04	Function code	04	Function code
0B B7	Register Start Addr	0B B7	Register Start Addr
00 02	2 registers	00 02	2 registers
03 C9	CRC Check	03 C9	CRC Check

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6.2. 03 function code

1) Acquire 1 setting

Send		Response	
01 03 0B B7 00 01 36 08		01 03 02 00 13 F9 89	
01	Address	01	Address
03	Function code	03	Function code
0B B7	Register Start Addr	0B B7	Register Start Addr
00 01	1 register	00 01	1 register
36 08	CRC Check	36 08	CRC Check

2) Acquire multiple settings

Send		Response	
01 03 0B B7 00 03 B7 C9		01 03 0C 00 13 00 02 00 14 AF 79	
01	Address	01	Address
03	Function code	03	Function code
0B B7	Register Start Addr	0C	Byte number
00 03	3 registers	00 13	Data (19year)
B7 C9	CRC Check	00 02	Data (2month)
		00 14	Data (20day)
		AF 79	CRC Check

6.3. 06 function code

Set 1 setting

Send		Response	
01 06 0B B7 00 13 7A 05		01 06 0B B7 00 13 7A 05	
01	Address	01	Address
06	Function code	06	Function code

0B B7	Address	0B B7	Address
00 13	data(19 year)	00 13	data(19 year)
7A 05	CRC Check	7A 05	CRC Check

6.4. 10 function code

Send		Response	
01 10 0B B7 00 03 06 00 13 00 02 00 14 51 65		01 10 0B B7 00 03 32 0A	
01	Address	01	Address
10	Function code	10	Function code
0B B7	Register Start Addr	0B B7	Register Start Addr
00 03	3 registers	00 03	3 registers
06	Byte number	32 0A	CRC Check
00 13	Data (19year)		
00 02	Data (2month)		
00 14	Data (20day)		
51 65	CRC Check		

6.5. Broadcast setting

Note: When the broadcast message uses the 06 or 10 function code for parameter setting, you only need to change the device address to FF. For the rest of the information, please refer to the 06 or 10 function code message example.

Send		Response	
FF 06 0B EB 1F 40 E7 C4		Slave responds, no reply message	
FF	Broadcast Address		
06	Function code		
0B EB	Register		
1F 40	Data (80kW)		
E7 C4	CRC Check		

Appendix 1 :

3044	Means		Display	
	2G Single-Phase	30KW Series 15kW Series ALL MII Series	2G Single-Phase	30KW Series 15kW Series ALL MII Series
0000H(0000)	Normal	Waiting	Generating	Waiting
0000H(0001)	\	Controlled Off Grid	\	Grid Off
0001H	\	OpenRun	\	OpenRun
0002H	Waiting	SoftRun	Waiting	SoftRun
0003H	Initializing	Generating	Initializing	Generating
1004H	Grid off	\	Grid Off	\
.....				
F010H	Grid surge(Warning)		Surge Alarm	
F011H	FAN fault (Warning)		Fan Alarm	
F013H	AC SPD ERROR(Warning)		VgSpdFail	
F014H	DC SPD ERROR (Warning)		DcSpdFail	
F015H	Fan fault (Warning External)		Fan_H Alarm	
F016H	Fuse Alarm(Warning)		FuseFail	
F017H	AC Phase Ground Fault(Warning)		L&PE FAIL	
.....				
1010H(0000)	Grid Over Voltage		OV-G-V	
1010H(0001)	Grid Over Voltage 01		OV-G-V01	
1010H(0002)	Grid Over Voltage 02		OV-G-V02	
1010H(0003)	Grid Over Voltage 03		OV-G-V03	
1010H(0004)	Grid Over Voltage 04		OV-G-V04	
1010H(0005)	Grid Over Voltage 05		OV-G-V05	
1011H(0000)	Grid Under Voltage		UN-G-V	
1011H(0001)	Grid Under Voltage 01		UN-G-V01	
1011H(0002)	Grid Under Voltage 02		UN-G-V02	

1012H(0000)	Grid Over Frequency	OV-G-F
1012H(0001)	Grid Over Frequency 01	OV-G-F01
1012H(0002)	Grid Over Frequency 02	OV-G-F02
1013H(0000)	Grid Under Frequency	UN-G-F
1013H(0001)	Grid Under Frequency 01	UN-G-F01
1013H(0002)	Grid Under Frequency 02	UN-G-F02
1014H	Grid reverse	Backfeed_lac
1015H	No Grid	NO-Grid
1016H	Grid Unbalance	G-PHASE
1017H	Grid Frequency Fluctuation	G-F-FLU

1018H	Grid Over Current	OV-G-I
1019H	Grid current tracking fault	IGFOL-F
101AH	Grid phase abnormal	PHASE-FAULT
101BH	Reserved	WORD1_B09
101CH	Igbt shift fault	IGBTShift-Pro
.....		
1020H(0000)	DC Over Voltage	OV-DC
1020H(0001)	DC Over Voltage 01	OV-DC01
1020H(0002)	DC Over Voltage 02	OV-DC02
1020H(0003)	PV Ground Fault	PVGndRun Fau
1020H(0004)	DCBoost Fault	BoostFail
1021H	DC Bus Over Voltage	OV-BUS
1022H	DC Bus Unbalance	UNB-BUS
1023H(0000)	DC Bus Under Voltage	UN-BUS
1023H(0001)	DC Bus Under Voltage 01	UN-BUS01
1023H(0002)	DC Bus Under Voltage 02	UN-BUS02
1024H	DC Bus Unbalance 2	UNB2-BUS
1025H	DC(Channel A) Over Current	OV-DCA-I
1026H	DC(Channel B) Over Current	OV-DCB-I
1027H	DC interference	DC-INTF.

1028H	DC reverse	Reve-DC
1029H	PV mid-point grounding	PvMidIso
102AH	The bus voltage is inconsistent (Note: the master and slave DSP bus voltage sampling is inconsistent)	Vbus-Sam
.....		
1030H	The Grid Interference Protection	GRID-INTF.
1031H	The DSP Initial Protection	INI-FAULT
1032H	Temperature Protection	OV-TEM01
1033H(0000)	PV Insulation fault	PV ISO-PRO
1033H(0001)	PV Insulation fault 01	PV ISO-PRO1
1033H(0002)	PV Insulation fault 02	PV ISO-PRO2
1034H(0000)	Leakage Current Protection	ILeak-PRO
1034H(0001)	Leakage Current Protection 01	ILeak-PRO01
1034H(0002)	Leakage Current Protection 02	ILeak-PRO02
1034H(0003)	Leakage Current Protection 03	ILeak-PRO03
1034H(0004)	Leakage Current Protection 04	ILeak-PRO04
1035H	Relay Protection	elayChk-FAIL
1036H(0000)	DSP_B Communication Fault	DSP-B-Com-Fau
1036H(0001)	DSP_B Sampling Fault	DSP-B-Sam-Fau

1037H	DC Injection Protection	DCInj-FAULT
1038H	12V Under Voltage Faulty	12Power-FAULT
1039H	Leakage Current Check Protection	ILeak-Check
103AH	Under temperature protection	UN-TEM
103BH	Reserved	WORD4_B00
103CH	Reserved	WORD4_B02
103DH	L&PE Fault	L&PE-FAULT
103EH	NTC Over Temp	OV-TEM02

.....		
1040H	AFCI Check Fault	AFCI-Check
1041H	AFCI Fault	ARC- FAULT
1046H	Grid INTF 02	GRID-INTF02
1047H	Grid current sampling error	IG-AD
1048H	IGBT over current	IGBT-OV-I
.....		
1050H	Instantaneous overcurrent of grid side current	OV-IgTr
1051H	Battery overvoltage hardware fault (hybrid)\DC bus hardware overvoltage (inverter)	OV-Vbatt-H(hybrid)\ OV-BUS-H(inverter)
1052H	LLC hardware overcurrent	OV-ILLC
1053H	Battery overvoltage detection	OV-Vbatt (hybrid) / PELD-HardLimit (Note: inverter)
1054H	Battery undervoltage detection	UN-Vbatt
1055H	battery not connected	NO-Battery
1056H	Bypass Overvoltage Fault	OV-VBackup
1057H	Bypass overload fault	Over-Load
1058H	DSP self-check exception	DspSelfChk
1059H	Vg sampling exception	Vg-Sample
105AH	DSP hardware mismatch	HardFault
105BH	DSP detects battery overcurrent	BAT-DOC
.....		
2010H	Fail Safe	Fail Safe
2011H	Fail Safe	Fail Safe
2012H	battery communication failure	CAN_Comm_FAIL
2014H	DSP communication failure	DSP_Comm_FAIL
2015H	BMS warning fault 01	Alarm1-BMS
2016H	Inconsistent battery	BatName-FAIL

	selection	
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2017H	BMS warning fault 02	Alarm2-BMS
2018H	DRM connection failed	DRM_LINK_FAIL
2019H	Abnormal meter selection	MET_SEL_FAIL
.....		
2020H	High ambient temperature for lead-acid batteries	HighTemp.AMB
2021H	Low ambient temperature for lead-acid batteries	LowTemp.AMB
.....		
2030H	Grid-connected Backup overload	BKAC Overload
.....		
2040H	PELD hard limit protection	PELD-HardLimit
.....		

Appendix 2:

3054 H	Standard			
	3PH (5-136K) (3PH Hybrid)	1PH (1PH Hybrid)	3PH(125K-1500V)	3PH(225K-1500V)
01H	G59/3	G59/3	G59/3	G59/3 (600V)
02H	UL480-13 (60Hz480V) (60Hz270V)	UL240-13	UL600-13	UL600-13
03H	VDE0126(380V)	VDE0126	VDE0126	VDE0126
04H	Reserve1	Reserve1	AS4777-15	AS4777-15 (Not Used)
05H	Reserve2	Reserve2	AS4777-02	AS4777-02 (Not Used)

06H	CQC-380A Note: 80-136K shows CQC-B-380A (B)	CQC (B)	CQC-600 (A)	CQC-800 (A)
07H	EN50438IE	EN50438IE	EN50438IE	ENEL (Not Used)
08H	UL-380V(60Hz 380V) Note : LV UL-220V(60H z220 V)	UL-208V	UL-380V(60Hz380V)	UL-380V(60Hz380V) (Not Used)
09H	MEX-CFE	MEX-CFE	MEX-CFE	MEX-800
0AH	User-define	User-define	User-define	User-define

0BH	VDE4105(380V)	VDE4105	VDE4105(380V)	VDE4105(380V) (Not Used)
0CH	DK1	DK1	EN50438DK	DK1
0DH	EN50549PO	EN50549PO	EN50549PO	EN50549PO
0EH	EN50549NL	EN50549NL	EN50549NL	EN50549NL
0FH	EN50549ES (Estonia)	EN50549ES	EN50549ES	EN50549ES
10H	EN50438L	EN50438L	EN50438L	EN50438L(800V)
11H	UL-480V-A	UL-240V-A	UL-600V-A	UL800-13
12H	UL-380V-A	UL-208V-A	UL-380V-A	UL-380V-A (Not Used)
13H	BRAZIL	BRAZIL	BRAZIL	BRAZIL
14H	Reserve3	Reserve3	AUS-Q-0.9	AUS-Q-0.9 (Not Used)
15H	Reserve4	Reserve4	AUS-Q-0.8	AUS-Q-0.8 (Not Used)
16H	G83/1	G83/1	G83/1	G83/1 (Not Used)
17H	RD1699	RD1699	RD1699	RD1699

				(Not Used)
18H	IEC61727	IEC61727	IEC61727	IEC61727
19H	GN-380L	G83/1-A	GN-600L	GN-800L
1AH	CQC-480A Note : 80-136K shows CQC-B-480A (B)	GNB	CQC-480V	CQC-480V (Not Used)
1BH	GN-HV-L	GNC	GN-HV-L	GN-HV-L (Not Used)
1CH	G59/3-A	NewZeal	G59/3-A	G59/3-A (600V)
1DH	4105/480(480V)	G83/3	4105/480(480V)	4105/480(480V) (Not Used)
1EH	Reserve5	Chile	AS4777_480	AS4777_480 (Not Used)
1FH	NewZeal	NRS097	NewZeal	NewZeal (Not Used)
20H	CQC-500A (B)	Philippin	CQC500	CQC500 (Not Used)
21H	CQC-540A Note : 80-136K shows CQC-B-540A (B)	N4105-BEL	CQC540	CQC540 (Not Used)
22H	GN-540L	IEC61727L	GN540L	GN540L

				(Not Used)
23H	N4105-BEL	KS1	N4105-BEL	N4105-BEL (Not Used)
24H	CHILE	France	CHILE	CHILE (Not Used)
25H	NRS097	ISONE240	NRS097	NRS097 (Not Used)

26H	GN380L-A	ISONE208	GN600L-A	GN800L-A
27H	GNHVL-A	ISONE240A	GNHVL-A	GNHVL-A (Not Used)
28H	NRS480	ISONE208A	NRS480	NRS480 (Not Used)
29H	CQC380DZ	GN300V	CQC600DZ	CQC800DZ
2AH	GN380DZL	MEA(THAILAN D)	GN600DZL	GN800DZL
2BH	ISONE480	R21P3-240	ISONE600	ISONE600
2CH	ISONE480A	R21P3-208	ISONE600A	ISONE800
2DH	KS1	R21P3-24A	KS1	KS4
2EH	R21P3-480	R21P3-20A	R21P3-600	R21P3-600
2FH	R21P3-48A	Sri Lanka	R21P3-60A	R21P3-800
30H	Philippin	PEA(THAILAN D)	Philippin	Philippin (Not Used)
31H	France	Reserve5	France	France (Not Used)
32H	Sri Lanka	Mala230LV	SRILANKA	SRILANKA
33H	MEA	Indon230V	THAILANDME A	THAILANDMEA (Not Used)
34H	PEA	G98	THAILANDPE A	THAILANDPEA (Not Used)
35H	Reserve6	G99	4777SA-48(480)	4777SA-48(480) (Not Used)
36H	Mala230LV	GEN50	Mala230LV	Mala230LV (Not Used)
37H	Mala277LV	GEN60	Mala277LV	Mala277LV (Not Used)
38H	Mala277MV	TW220(TAIWA N)	Mala277MV	Mala277MV (Not Used)
39H	Indon230V	TW110(TAIWA N)	Indon230V	Indon230V (Not Used)
3AH	Dubai-A (Dubai MV)	DK230V	DEWA230LV	DEWA230LV (Not Used)

3BH	Dubai-B (Dubai LV)	Barbados	DEWA277LV	DEWA277LV (Not Used)
3CH	DEWA277MV	BRAZIL-H	DEWA277MV	DEWA277MV

				(Not Used)
3DH	G98	G99-N	G98	G98
3EH	G99	CEI 0-21(Italy)	G99	G99
3FH	BDEW230V	MEX-220V(M EX-11 0V Note : LV Display)	BDEW230V	BDEW230V (Not Used)
40H	BDEW277V	MEX220-A(M EX110 -A Note : LV Display)	BDEW277V	BDEW277V (Not Used)
41H	GEN50	SG1	GEN50	GEN50 (Not Used)
42H	GEN60	Reserve6	GEN60	GEN60 (Not Used)
43H	Reserve7	Reserve7	4777SA-40(380)	4777SA-40(380) (Not Used)
44H	KS2(Korean Non-standard)	EN50549	KS2(Korean Non-standard)	KS2(Korean Non-standard) (Not Used)
45H	TW220(TAIWA N)	PH-L(Philippin)	TW220(TAIWA N)	TW220(TAIWAN) (Not Used)
46H	DK277V	C10/11(比利时)	DK277V	DK277V (Not Used)
47H	DK230V	DK2	DK230V	DK230V (Not Used)
48H	Barbados	G98-NI	Barbados	Barbados (Not Used)
49H	IEC61727L	G99-NI	IEC61727L	IEC61727L
4AH	SG1	IRAN	SG1	SG1

4BH	G99N	EIFS-SW(Sweden)	G99-N	G99-N
4CH	MEX-480V (MEX-220V Note: LV Display)	R14-240A (Hawaii)	MEX-480V	MEX-480V (Not Used)
4DH	MEX480V-A (MEX220V-A Note: LV Display)	R14-208A (Hawaii)	MEX480-A	MEX480-A (Not Used)
4EH	Reserve8	TOR (Austria)	4777WA-40(380)	4777-A-EH
4FH	Reserve9	R14-240 (Hawaii)	4777WA-48(480)	4777-B-EH
50H	Reserve10	R14-208 (Hawaii)	4777NW-40(380)	4777-C-EH
51H	Reserve11	Reserve8	4777NW-48(480)	4777-N-EH
52H	EN50549	GREECE	EN50549L	EN50549L

53H	CEI 0-21(Italy)	HK230(HongKong)	CEI 0-21(Italy)	CEI 0-21(Italy)
54H	PH-L(Philippin)	RENBLAD(Norway342)	PH-L(Philippin)	PH-L(Philippin) (Not Used)
55H	C10/11(Belgium)	4777-A (AS4777-2020A)	C10/11(Belgium)	C10/11(Belgium)
56H	DK2	4777-B	DK2	DK2
57H	G98-NI	4777-C	G98-NI	G98-NI
58H	G99-NI	4777-N (New Zealand)	G99-NI	G99-NI

59H	IRAN	NTS631 (Reserve9 description: This display is only for grid-connecte d with LCD display, others remain unchanged)	Iran	Iran
5AH	EIFS-SW(Swede n)	UL-240-18 (Note: UL-1574 A is old UL1547-2014 A2 is new UL1547-2018)	EIFS-SW(Swede n)	EIFS-SW(Sweden)
5BH	KS3	UL-208-18	EN50549-2(600V)	EN50549-2(800V)
5CH	TOR(Austria)	EN50549IE	CEA600(India)	CEA800(India)
5DH	BRAZIL-H	VIETNAM	Puerto600	Puerto600
5EH	CQC-A-380 (Only for 80-110K/90-1 36K (A))	GNSD	BRAZIL-H	SG-800V (Singapore)
5FH	CQC-A-480 (Only for 80-110K/90-1 36K (A))	PN-50549 (Poland)	VIETNAM	G99-B
60H	CQC-A-540 (Only for 80-110K/90-1 36K (A))	ESB-Micro (Note: New Ireland EN50549)	VDE4110	GREECE800
61H	G99-B	ESB-Mini (Note: New Ireland		NTS631(800V)

		EN50549)		
62H	Reserve12	Israel		VIETNAM
63H	Reserve13	R14H-240 (Hawaii)		VDE4110
64H	GREECE	R14H-208 (Hawaii)		UL800-18 (Note : UL-1574 A is old UL1547-2014 A2 is new UL1547-2018)
65H	HK230V			UL600-18 (Note : UL-1574 A is old UL1547-2014 A2 is new UL1547-2018)
66H	REN342 (RENBLAD Norway342)			PN-50549 (Poland)
67H	CEI 0-16			
68H	NTS631 (Reserve14 description: This display is only for grid-connected u with LCD display, others remain unchanged)			
69H	4777-A (AS477-2020)			
6AH	4777-B			
6BH	4777-C			
6CH	4777-N (NEW ZEALAND)			

6DH	4777-A-H			
6EH	4777-B-H			
6FH	4777-C-H			
70H	4777-N-H			
71H	UL480-18 (Note: UL-1574 A is old UL1547-2014			

	A2 is new UL1547-2018)			
72H	EN50549IE			
73H	EN50549-2			
74H	VIETNAM			
75H	VDE4110			
76H	GNSD-A380			
77H	GNSD-A-HV			
78H	GNSD-A540			
79H	GNSD-B380			
7AH	GNSD-B-HV			
7BH	GNSD-B540			
7CH	PN-50549(Pol and)			
7DH	ESB-Micro (Note: New Ireland EN50549)			
7EH	ESB-Mini (Note: New Ireland EN50549)			

7FH	Israel(以色列)			
80H	Egypt (埃及)			
81H	KS3-LVRT			
82H	CQC3310 (Note: dedicated for hybrid, reserved for grid-connecte d machines)			